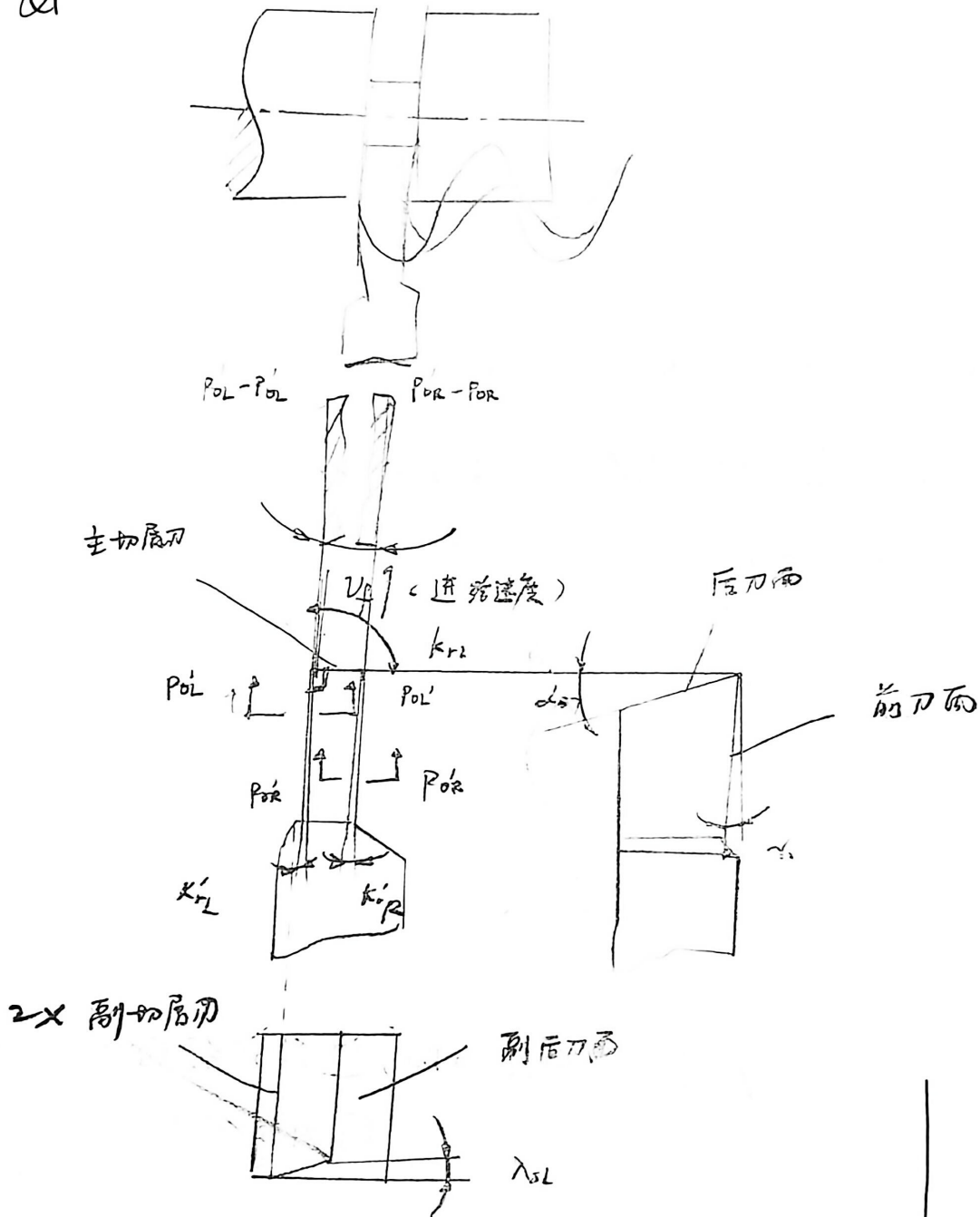


潘浩强 20200404 数控 制造 HW

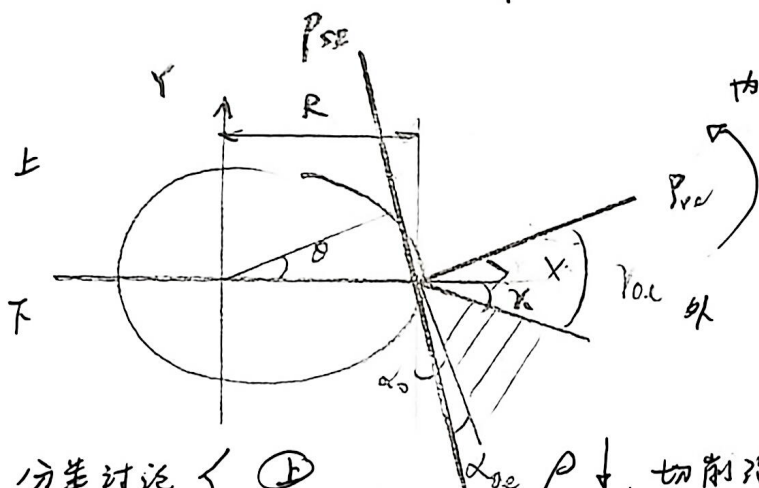
Q1



$\gamma_0 = 10; \quad 2\alpha = 5; \quad 5 = \alpha_0$

$$\gamma_0 = 10^\circ, \quad \alpha_0 = 5^\circ, \quad \delta = \alpha_0$$

题意: ① 刀具工作角度变化, ② 标准角度 ③ $\alpha = 0^\circ$, 位置
④ 采取控制一致工作角



分类讨论 { 上
下

$\rho \downarrow$, 切削沿切向, 进给沿径向指向曲线内侧
同内部倾斜, 前角 $\uparrow (\alpha_{or})$
后角 $\downarrow (\alpha_{os})$

$$\tan \beta = \frac{v_f}{v_c} = \frac{-\frac{dv}{dt}}{\frac{d\theta}{dt}} = \frac{-d(R - CO)}{(R - CO)d\theta} = \frac{C}{R - CO}$$

(工作前角) $\gamma_{oe} = \gamma_o + \eta$

(工作后) $\alpha_c = \alpha_0 - \eta$

$$\therefore 0 = 5 - \eta \quad \eta = 5^\circ$$

$$\tan \theta = \frac{3.2}{45 - 3.20}, \quad \theta = 2.63 \text{ rad} = 151^\circ$$

